

PHY5503: INTRODUCTION TO QUANTUM TECHNOLOGY

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Introduction to Quantum Technology

Subject Code

PHY - Physics

Course Number

5503

Academic Unit

Physics (PHY)

College/School

College of Science (SI)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

PHY8513 Introduction to Quantum Technology

Part II Course Details

Abstract

The Introduction to Quantum Technology course offers a captivating journey into the fascinating world of quantum mechanics and its ground-breaking applications. This course provides a comprehensive overview of quantum technology's fundamental principles and potential impact. Students will study quantum information science, exploring concepts such as qubits, quantum gates, and entanglement. They will uncover the mysteries of quantum computing, discovering powerful algorithms and the intricacies of quantum error correction. The course also covers quantum communication, cryptography, and secure communication protocols. Moreover, students will explore quantum sensing and metrology, unlocking the potential for precise measurements and imaging. Upon completion, students will possess a solid foundation in quantum technology, empowering them to pursue further studies or careers in this rapidly advancing field that promises to revolutionize industries worldwide.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Understand various quantum enabled technologies		x	x	
2	Understand Quantum superposition, entanglement measurement		x	x	
3	Understand Qubits and quantum states		x	x	
4	Understand Quantum gates and circuits		x	x	
5	Understand different physical platforms		x	x	
6	Understand Quantum algorithms (e.g., Shor's algorithm, Grover's algorithm)		x	x	
7	Understand Quantum cryptography and secure communication		x	x	
8	Understand Quantum key distribution protocols		x	x	
9	Understand Quantum teleportation and quantum networks		x	x	
10	Understand Quantum-enhanced measurements		x	x	
11	Understand Applications in precision measurement and imaging		x	x	
12	Understand Quantum simulators and their applications		x	x	

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	Presentation of course material	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12	3

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- for nil entry)	Allow Use of GenAI?
1	Assignments	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12	40	-	Yes
2	Test	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12	30	-	Yes

Continuous Assessment (%)

70

Examination (%)

30

Examination Duration (Hours)

2

Minimum Continuous Assessment Passing Requirement (%)

30

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination and continuous assessment must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Tests

Criterion

The student demonstrates an understanding of the principles of quantum physics for solving common quantum technology problems.

Excellent (A+, A, A-)

High (Outstanding achievement and accurate understanding)

Good (B+, B, B-)

Significant (Good achievement with largely accurate understanding)

Fair (C+, C, C-)

Satisfied (Moderate achievement with some accurate understanding)

Marginal (D)

Basic (Essential achievement with a basic understanding)

Failure (F)

Not reaching marginal level

Assessment Task

Assignments

Criterion

The student completes all the assignments and demonstrates a good understanding of the taught material by solving the given problems.

Excellent (A+, A, A-)

High (Outstanding achievement and accurate understanding)

Good (B+, B, B-)

Significant (Good achievement with largely accurate understanding)

Fair (C+, C, C-)

Satisfied (Moderate achievement with some accurate understanding)

Marginal (D)

Basic (Essential achievement with a basic understanding)

Failure (F)

Not reaching marginal level

Assessment Task

Examination

Criterion

The student demonstrates an understanding of the principles of quantum physics for solving common quantum technology problems.

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Failure (F)

Not reaching marginal level

Part III Other Information

Keyword Syllabus

- Introduction to Quantum Mechanics
 - Wave-particle duality, uncertainty principle, quantum states, operators, measurement, observables

- Quantum Computing
Qubits, quantum gates, quantum algorithms, Grover's algorithm, Shor's algorithm, quantum simulation, optimization, error correction, fault tolerance
- Quantum Communication
Quantum key distribution, QKD, quantum teleportation, quantum networks, secure communication, quantum internet
- Quantum Sensing
Quantum sensing, quantum metrology, precision measurements
- Technologies
Sensing, measuring, imaging, communication, simulation and computing
- Platforms
Superconducting qubits, Trapped ions, photonics, Nuclear magnetic resonance, Quantum dots, Diamond vacancies.

Reading List

Compulsory Readings

Title	
1	Michael A. Nielsen, Isaac L. Chuang Quantum Computation and Quantum Information CUP 2010. https://doi.org/10.1017/CBO9780511976667

Additional Readings

Title	
1	R. Loudon, Quantum Theory of Light, 3rd Edition (Oxford University Press, 2000)